

Theory of Automata

Assignment # 4

Start Date: 06/01/2025

Section: B

Total Marks: 35

Due Date: 11/01/2025

Program: BSCS

Q1. Give the CFG for the following Languages

- I. All words that don't end at double letter.
- II. $\{W \mid \text{the length of } w \text{ is odd}\}$
- III. $\{W \mid w \text{ contains more } a\text{'s than } b\text{'s}\}$
- IV. $a^n b^m c^m$ where $n, m = 0, 1, 2, 3, \dots$
- V. $a^n b^n c^m d^{2m}$ where $n, m = 0, 1, 2, 3, \dots$

Q2. Convert the following CFG into CNF

$$S \rightarrow aXbX$$

$$X \rightarrow aY \mid bY \mid \epsilon$$

$$Y \rightarrow X \mid c$$